

TrackMind AI - Total System Analysis

AI-Powered Automatic Block Planning for Indian Railways (SIH 26027)

Architecture, Full Technology Stack, Optimization Algorithms & Multi-Agent Network

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1. Executive Overview & Problem Statement

In Indian Railways, three core infrastructure departments constantly require track maintenance blocks (temporary corridor closures):

- **Engineering (Track / P-Way):** Rail fractures, deep screening, track packing, rail renewal (tracked via TMS).
- **Signal & Telecom (S&T):** Point machines, track circuits, axle counters, signals (tracked via SMMS).
- **Traction Distribution (TRD):** OHE wire wear, catenary maintenance, power isolations (tracked via TDMS).

Historically, these departments requested blocks in silos through BDMS with zero real-time cross-departmental coordination. This resulted in scheduling collisions, severe passenger train delays, and unexploited corridor windows. TrackMind AI solves this by unifying maintenance requests, train timetables, and goods traffic forecasts into an automated mathematical optimization and multi-agent AI system.

2. Comprehensive Technology Stack Architecture

Layer	Technologies Used	Role & Operational Implementation
Frontend UI	Streamlit (1.38+), Custom CSS3	Reactive full-stack web dashboard with dark glassmorphic styling and state management.
Geospatial GIS	Folium (0.20+), Leaflet.js	Satellite multi-track map, station pill markers (DivIcon), live train positions & telemetry.
Data Visualization	Plotly Express & Graph Objects	Interactive corridor Gantt timelines, capacity heatmaps, KPI meters, and defect charts.
Voice & Audio	Web Speech API (STT / TTS)	Browser-native multilingual speech recognition & synthesis (English, Hindi, Telugu).
Operations Research	Google OR-Tools CP-SAT	Constraint Programming engine ensuring zero collisions and timetable conflict avoidance.
Machine Learning	Scikit-Learn (RandomForest, Isolation)	Failure risk prediction (RandomForest) and track anomaly defect clustering (IsolationForest).
Data Processing	Pandas (2.2+), NumPy, Faker	Vectorized DataFrame transformations and 2,000+ realistic synthetic railway records.
GenAI & LLM	Groq SDK (LLaMA-3.3-70B)	Sub-second AI reasoning with automated fallback to LLaMA-3.1-8B, Mixtral, and Gemma2.
Knowledge RAG	Custom NL-to-SQL + Vector RAG	Live database querying engine against railway.db for zero-hallucination factual stats.
Database & Storage	SQLite3 (WAL Mode)	Relational database with Write-Ahead Logging and 30s busy timeout for concurrent safety.
Security & RBAC	Bcrypt (4.2+)	Salted password encryption and Role-Based Access Control (Admin, Eng, S&T, TRD).
Reporting Engine	FPDF2 (2.8+)	Automated server-side generation of downloadable maintenance block PDF schedules.
DevOps & Deploy	GitHub, Streamlit Cloud, Python 3.13	Continuous deployment with pinned runtime and automated dependency management.

3. Mathematical Optimization Engine (Google OR-Tools CP-SAT)

The scheduling core (scripts/optimizer.py) formulates maintenance planning as a binary integer constraint satisfaction and optimization problem (CP-SAT):

- **Hard Constraint 1 (No Collisions):** Each corridor time slot can hold at most one maintenance block unless merged into an approved shadow block.
- **Hard Constraint 2 (Train Timetable Protection):** Slots overlapping scheduled passenger train departures on the same section are purged via `filter_slots_against_timetable()`.
- **Hard Constraint 3 (Section & Duration Matching):** Tasks can only be scheduled into slots on their identical section ID with `slot duration >= estimated task duration`.
- **Objective Function:** Maximize cumulative Priority Score of all scheduled tasks across 7-day weekly and 30-day monthly rolling horizons.

4. Predictive Risk & Defect Scoring (ML Layer)

Every maintenance defect across TMS, SMMS, and TDMS undergoes dual scoring before reaching the solver

(scripts/scoring_models.py):

- **Priority Score (0-100):** Severity Weight (Critical=40, High=25, Medium=12, Low=5) + Overdue Factor (up to 30 pts) + Trains Affected Impact (up to 30 pts).
- **Failure Risk Score (RandomForest):** Trained on historical failure features to compute the probability (0-100%) of in-service failure before scheduled repair.
- **Final Priority Blending:** Final Priority = $0.60 * \text{Priority Score} + 0.40 * \text{Failure Risk Score}$.
- **Anomaly Detection (IsolationForest):** Detects abnormal defect clustering per track section, automatically alerting controllers to systemic track degradation.

5. 32 Autonomous Domain Agents

The system deploys 32 specialized autonomous Python agents (scripts/agents.py) organized into functional clusters:

- **Core Coordination:** CoordinatorAgent, ReplanningAgent, TrafficAgent, DepartmentAgent (proposes departmental blocks).
- **Safety & Compliance:** ComplianceAgent (SLA and section conflict clustering), SafetyClearanceAgent (G&SR rules), AnomalyDetectionAgent.
- **Locopilot & Speed:** LocopilotSpeedAgent, TelemetrySimulatorAgent, DynamicHeadwayAgent, SingleLineWorkingAgent, TSRLifecycleAgent.
- **Resource & Freight:** TrackMachinePackerAgent, CrewHOERAgent (Hours of Employment Regulations), TractionAwareRouterAgent, FOISDemurrageAgent.
- **Passenger & Dispatch:** PassengerAdvisoryAgent (automated delay bulletins), CostOptimizationAgent, FeedbackLoopAgent.

6. Multilingual Generative AI & Voice Assistant

Integrated in app/chatbot.py for hands-free and natural language interaction:

- **Groq LLaMA-3.3-70B API:** Sub-second reasoning with strict 4-second timeout and multi-model fallback (LLaMA-3.1-8B, Mixtral, Gemma).
- **Dynamic NL-to-SQL RAG:** Converts plain English/Hindi/Telugu queries into verified live SQLite SELECT queries against railway.db.
- **Voice STT & TTS:** Web Speech API integration supporting English (en-IN), Hindi (hi-IN), and Telugu (te-IN) natively in-browser.
- **Zero Hallucination Guard:** Rigid domain prompting ensures the model never invents metrics, citing railway.db as the source of truth.

7. Geospatial Multi-Division Corridors

Dynamic satellite mapping and Gantt timelines support 4 prominent Indian Railways divisions:

- **Khurda Road (KUR):** East Coast Railway (ECoR) - Main Trunk Cuttack-Bhubaneswar-Puri-Brahmapur network.
- **Vijayawada (BZA):** South Central Railway (SCR) - High-density trunk route connecting Rajahmundry-Vijayawada-Tenali.
- **Secunderabad (SC):** South Central Railway (SCR) - Kazipet-Secunderabad-Vikarabad passenger/freight crossroads.
- **Howrah (HWH):** Eastern Railway (ER) - Suburban and mainline high-frequency corridor.

8. Database Reliability & Security

- **SQLite WAL Mode:** Enabled PRAGMA journal_mode=WAL and PRAGMA busy_timeout=30000 for concurrent non-blocking reads and writes.
- **Role-Based Access Control:** Bcrypt password hashing with strict separation between Central Controller and Engineering/S&T/TRD portals.
- **Department Alert Isolation:** Department users receive only role-specific alerts (e.g. S&T never sees OHE electrical notifications).

9. Business & Operational Impact on Indian Railways

- **100% Conflict Elimination:** Zero departmental clashes through mathematical constraint satisfaction.
- **Punctuality Maximization:** Train timetable departure protection eliminates passenger train detentions during maintenance windows.
- **Capacity Expansion:** Shadow block clustering allows Track, S&T, and TRD to work in unified windows, saving up to 35%.

corridor hours.

- **Safety Governance:** Automatic G&SR safety certificate generation ensures full statutory compliance before power restoration.